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Experiment 3: Water Jug Problem (Python)

Aim

To write a Python program to solve the **Water Jug Problem**.

Algorithm

1. Take the capacities of two jugs.
2. Fill the first jug.
3. Pour water from the first jug to the second jug until it is full.
4. Empty the second jug if it becomes full.
5. Repeat the process until the required amount of water is obtained.
6. Display the steps.

Program

```
a = 4

b = 3

target = 2


x = 0

y = 0


while x != target and y != target:

  if x == 0:

    x = a

    print(x, y)

  elif y == b:

    y = 0

    print(x, y)

  else:

    t = min(x, b - y)

    x -= t
```

```
    y += t
    print(x, y)

print("Target reached")
```

Output

```
4 0
1 3
1 0
0 1
4 1
2 3
Target reached
```

Result

Thus, the Python program to solve the Water Jug Problem was executed successfully, and the required amount of water (2 litres) was obtained.